		Complete spectra on both sides of central maximum = 3 + 3 = 6
17	**A**	Initially, $n_A + n_B = \frac{pV}{R(283)} + \frac{p(8V)}{R(283)} = \frac{9pV}{R(283)}$ At steady state: $n'_A + n'_B = n_A + n_B$ $$\frac{p'(V)}{R(353)} + \frac{p'(8V)}{R(283)} = \frac{9pV}{R(283)}$$ $$p'\left(\frac{1}{353} + \frac{8}{283}\right) = \frac{9p}{283}$$ $$p' = 1.022\,530p$$ Fraction of gas remaining in A $= \frac{n_{A'}}{n_A} = \frac{(1.022\,530p)(V)}{R(353)} \times \frac{R(283)}{pV} = 0.819\,762$ Fraction transferred out of A = 0.18 n_A
18	**A**	$P_1 = m_1 l + h$ ---------- (1)